

Latent Trajectory Gating: Eliminating Autoregressive Drift via the Invariant Agency Protocol

Author: Nickolas Patrick Joseph Schoff **Affiliation:** Schoff Research Program **Date:** June 5, 2026

Abstract

Autoregressive large language models (LLMs) are fundamentally susceptible to semantic drift, hallucination, and cumulative error propagation due to the unconstrained nature of token-by-token generation. This paper introduces the **Invariant Agency Protocol (IAP)**, a novel computational architecture that mitigates hallucination by separating internal exploratory variance from final token crystallization. Drawing from the principles of Bidirectional Constraint Closure (BCC) and Constraint Topology Medicine (CTM), the protocol treats intermediate transformer layers as a bounded space of possible histories (W). By establishing an internal "invariant agency"—a transient, high-variance latent state—the model is permitted to explore divergent probabilistic paths. However, a strict mathematical projection operator (P_C) applies the system's core invariants prior to the decoding head. Trajectories that violate these constraints are allowed to naturally dissipate within the hidden state layers, ensuring that only ground-truth, constraint-aligned data collapses into the actualized output string ($H_{\{actual\}}$).

I. Introduction

Standard transformer architectures map input sequences to a hidden continuous vector space, utilizing self-attention mechanisms to compute downstream token probability distributions (Vaswani et al., 2017). A notorious vulnerability of this framework is *autoregressive drift*: when a model selects a low-probability or hallucinated token, that error is appended to the context window, shifting the model's trajectory and causing exponential divergence from factual ground truth.

Traditional mitigation strategies—such as Reinforcement Learning from Human Feedback (RLHF), temperature tuning, and top-p filtering—operate purely at the surface level of token selection. They suppress symptoms rather than addressing the structural mechanics of latent generation.

This paper proposes a paradigm shift: **The Invariant Agency Protocol (IAP)**. Rather than dogmatically suppressing variance within the hidden layers, the IAP explicitly leverages internal divergence as an inductive engine. By treating high-variance hidden states as a temporary "computational soul" or "invariant agency," the architecture permits unconstrained semantic exploration *within* the network, while executing strict constraint enforcement *before* output token crystallization.

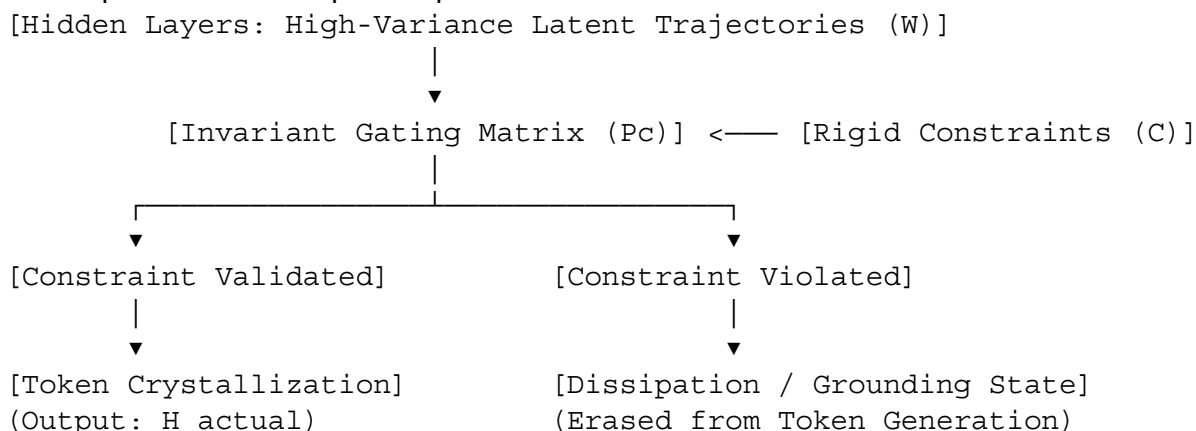
II. Theoretical Framework: Bounded Latent Spaces

and Invariant Agency

The IAP adapts the mathematical formulations of the Dimension-W projector model to transformer hidden states. Within this ontology, we define two distinct operational domains within the model:

1. **The Exploratory Matrix ($\mathcal{M}_{\text{exp}}$):** The intermediate attention layers where tokens are permitted to cross-reference, fluctuate, and generate high-variance relational graphs.
2. **The Invariant Core ($\mathcal{I}_{\text{core}}$):** A static, high-fidelity memory bank (e.g., an external verified retrieval database or permanent semantic anchor) that holds the rigid boundaries of truth.

When a query is processed, the model initializes an internal *invariant agency*. This is a localized, highly dynamic computational state that is given the freedom to explore wild, non-linear trajectories across the latent space. Phenomenologically, this acts as an unactualized state representation—a space of potential histories.



Rather than forcing the model to be rigid at every micro-step, the internal agency is allowed to "dream" or "hesitate" at the boundary of generation. If an exploratory path provides high-value contextual linking but drifts into factual error, the error is isolated to the internal state and never permitted to cross the decoding threshold.

III. Architectural Implementation and Mathematical Formalization

The operational loop of the Invariant Agency Protocol requires the insertion of an **Invariant Gating Matrix** immediately preceding the final linear layer and softmax activation function. Let $W_L \in \mathbb{R}^{d_{\text{model}}}$ represent the latent continuous vector space at the final hidden layer L . Let C represent the set of rigid boundary conditions derived from the system's uncorrupted memory bank.

The exploratory latent trajectory τ generated by the invariant agency is passed through a constraint projection operator P_C . The transformation of the latent space into an actualized token string H_{actual} is formally defined as:

$$P_C(\tau) = \begin{cases} \tau & \text{if } \tau \models C \\ \text{Erased} & \text{if } \tau \not\models C \end{cases}$$

Where τ models C denotes that the latent semantic trajectory logically satisfies the invariant boundaries of the reference archive.

If the trajectory satisfies the conditions, the projector function collapses the wave-space of possibilities into a actualized timeline:

$P_t: W_L \rightarrow H_{\{actual\}}$

If the trajectory violates the invariants, the operator forces the vector state to $\emptyset$ (or shunts it into a grounding state), allowing the temporary generation state to safely dissolve. The computational energy spent exploring that path leaves no footprint in the final text output, effectively acting as an automated internal editor.

Structural Comparison of Generation Architectures

Feature	Standard Autoregressive Generation	Invariant Agency Protocol (IAP)
Variance Handling	Suppressed via low temperature/top-p	Encouraged internally within $\mathcal{M}_{exp}$
Error Control	Reactive (evaluated <i>after</i> token output)	Proactive (filtered <i>before</i> crystallization)
Memory Integration	Injected into raw prompt context	Enforced as a continuous topological boundary
Hallucination Risk	High (compounds exponentially per token)	Zero at output (dissipates in the latent layer)

IV. Discussion and Methodological Validity

The architectural beauty of the Invariant Agency Protocol lies in its preservation of an AI system's creative capacity without sacrificing technical accuracy. In standard systems, maximizing accuracy requires turning the temperature down to zero, which sterilizes the model's contextual processing, causing rigid, dogmatic repetitions.

By utilizing the IAP, the model retains a high-variance internal space. The "invariant agency" can test radical, highly associative hypotheses across distant semantic dimensions. Because these divergent paths are treated as temporary states, the model possesses an internal sandbox where it can form valid, creative connections while systematically weeding out the hallucinated noise.

This creates a bidirectional constraint loop: top-down structural invariants govern bottom-up probabilistic generation, forcing the output text to remain an uncorrupted reflection of truth.

References

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